

Variance Reduction Note 4: The Convergence of SVRG

Alina Lodhi

Abstract

Stochastic variance-reduced gradient (SVRG) uses a full gradient at a snapshot point to correct inexpensive component gradients. This fourth expository note on variance reduction explains that control variate and gives a self-contained proof of linear convergence for the classical strongly convex finite-sum setting. Each component is assumed convex and smooth, while only their average must be strongly convex. A second-moment bound links the estimator to the objective gaps at the current iterate and the snapshot. Summing a distance recursion over an epoch then yields a geometric decrease and an oracle complexity of order $(n + L/\mu) \log(1/\tau)$ for objective accuracy τ , with the initial gap fixed. The result uses a random snapshot update and counts all full-gradient computations. No new algorithm or rate is claimed.

1 A snapshot as a control variate

Consider the finite-sum problem

$$\min_{x \in \mathbb{R}^d} F(x) := \frac{1}{n} \sum_{i=1}^n f_i(x). \quad (1)$$

Sampling a single component gradient is inexpensive, but its noise can persist at the minimizer: individual $\nabla f_i(x_*)$ need not vanish even when $\nabla F(x_*) = 0$. SVRG, introduced by Johnson and Zhang [6], corrects this noise using a snapshot $\tilde{x}$ and its exact gradient. For a uniform random index i , the estimator is

$$v(x, \tilde{x}; i) = \nabla f_i(x) - \nabla f_i(\tilde{x}) + \nabla F(\tilde{x}). \quad (2)$$

The subtracted and added terms have the same expectation. Consequently, the estimator is unbiased for $\nabla F(x)$, while the shared index cancels much of the component noise. At $x = \tilde{x}$, it equals the full gradient for every index.

This note develops the classical strongly convex analysis. Related finite-sum methods include SAG, SAGA, and proximal SVRG [4, 8, 10]. Recursive approaches such as SARAH, SPIDER, and PAGE [23, 26, 31] use a different form of correction. An overview of these method families appears in [2].

2 Assumptions and the epoch algorithm

Assumption 1. Each $f_i : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex, differentiable, and has an L -Lipschitz gradient, with $L > 0$. Their average F is μ -strongly convex for some $0 < \mu \leq L$: for all x, y ,

$$F(y) \geq F(x) + \langle \nabla F(x), y - x \rangle + \frac{\mu}{2} \|y - x\|^2. \quad (3)$$

Let x_* be its unique minimizer, $F_* = F(x_*)$, and $h(x) = F(x) - F_*$. Norms are Euclidean.

The components need not be strongly convex and need not share a minimizer. We do not assume uniformly bounded stochastic-gradient variance. The proof instead derives a bound that shrinks with the objective gaps.

Fix an initial snapshot $\tilde{x}_0$, a step size $\eta > 0$, and an integer epoch length $m \geq 1$. At epoch $s = 0, 1, \dots$:

1. Compute the full snapshot gradient $\tilde{g}_s = n^{-1} \sum_{i=1}^n \nabla f_i(\tilde{x}_s)$, and initialize $x_0 = \tilde{x}_s$.
2. For $t = 0, \dots, m-1$, draw a fresh independent uniform index $i_t \in \{1, \dots, n\}$ and update

$$v_t = \nabla f_{i_t}(x_t) - \nabla f_{i_t}(\tilde{x}_s) + \tilde{g}_s, \quad x_{t+1} = x_t - \eta v_t. \quad (4)$$

3. Choose J_s independently and uniformly from $\{0, \dots, m-1\}$, and set $\tilde{x}_{s+1} = x_{J_s}$.

The inner index t restarts at each epoch. This random snapshot choice is the variant analyzed below; the proof averages the function values of precisely these m iterates. An implementation may draw J_s in advance and retain its selected iterate without storing the whole trajectory. The cost per epoch is at most $n + 2m$ component-gradient evaluations, using $O(d)$ auxiliary vector storage when snapshot component gradients are recomputed as needed.

3 A short linear-convergence proof

The proof first controls component-gradient differences at the solution, then bounds the SVRG estimator, and finally sums a distance recursion over one epoch.

Lemma 1 (Gradient differences and objective gaps). *Under Assumption 1, for every x ,*

$$\frac{1}{n} \sum_{i=1}^n \|\nabla f_i(x) - \nabla f_i(x_\star)\|^2 \leq 2Lh(x). \quad (5)$$

Proof. Define $\psi_i(u) = f_i(u) - f_i(x_\star) - \langle \nabla f_i(x_\star), u - x_\star \rangle$. Convexity gives $\psi_i(u) \geq 0$, and ψ_i is L -smooth. Apply its descent inequality at $u = x$ with $y = x - \nabla \psi_i(x)/L$:

$$0 \leq \psi_i(y) \leq \psi_i(x) - \frac{1}{2L} \|\nabla \psi_i(x)\|^2.$$

Thus $\|\nabla f_i(x) - \nabla f_i(x_\star)\|^2 \leq 2L\psi_i(x)$. Averaging over i gives (5), because $n^{-1} \sum_i \nabla f_i(x_\star) = \nabla F(x_\star) = 0$. $\square$

Lemma 2 (Unbiasedness and second moment). *For fixed $x, \tilde{x}$ and a uniform index i , the estimator in (2) satisfies*

$$\mathbb{E}_i[v] = \nabla F(x), \quad \mathbb{E}_i\|v\|^2 \leq 4L[h(x) + h(\tilde{x})]. \quad (6)$$

Proof. Unbiasedness follows by averaging (2). Set $a_i = \nabla f_i(x) - \nabla f_i(x_\star)$ and $c_i = \nabla f_i(\tilde{x}) - \nabla f_i(x_\star)$. Then $v = a_i - (c_i - \mathbb{E}_i c_i)$. Hence

$$\begin{aligned} \mathbb{E}_i\|v\|^2 &\leq 2\mathbb{E}_i\|a_i\|^2 + 2\mathbb{E}_i\|c_i - \mathbb{E}_i c_i\|^2 \\ &\leq 2\mathbb{E}_i\|a_i\|^2 + 2\mathbb{E}_i\|c_i\|^2 \leq 4L[h(x) + h(\tilde{x})], \end{aligned}$$

using the variance identity and Lemma 1. $\square$

Theorem 1 (Contraction per epoch). *Suppose Assumption 1 holds. If $0 < \eta < 1/(2L)$ and m are chosen so that*

$$\rho := \frac{1}{\mu\eta m(1-2L\eta)} + \frac{2L\eta}{1-2L\eta} < 1, \quad (7)$$

then after S epochs,

$$\mathbb{E}h(\tilde{x}_S) \leq \rho^S h(\tilde{x}_0). \quad (8)$$

In particular, $\eta = 1/(10L)$ and $m = \lceil 50L/\mu \rceil$ ensure $\rho \leq 1/2$.

Proof. Condition on the start of an epoch and denote its snapshot by $\tilde{x}$. All expectations in this paragraph are conditional on that snapshot. Condition further on the history before drawing i_t . Unbiasedness, convexity of F , and Lemma 2 imply

$$\begin{aligned}\mathbb{E}_{i_t} \|x_{t+1} - x_\star\|^2 &= \|x_t - x_\star\|^2 - 2\eta \langle \nabla F(x_t), x_t - x_\star \rangle + \eta^2 \mathbb{E}_{i_t} \|v_t\|^2 \\ &\leq \|x_t - x_\star\|^2 - 2\eta(1 - 2L\eta)h(x_t) + 4L\eta^2 h(\tilde{x}).\end{aligned}$$

Take expectations over the inner history and sum for $t = 0, \dots, m-1$. The squared distances telescope, and the final one is nonnegative:

$$2\eta(1 - 2L\eta) \sum_{t=0}^{m-1} \mathbb{E} h(x_t) \leq \|\tilde{x} - x_\star\|^2 + 4L\eta^2 m h(\tilde{x}).$$

Strong convexity gives $\|\tilde{x} - x_\star\|^2 \leq 2h(\tilde{x})/\mu$. Divide by $2\eta(1 - 2L\eta)m$ and use the independent uniform snapshot choice to obtain $\mathbb{E}[h(\tilde{x}_{s+1}) \mid \tilde{x}_s] \leq \rho h(\tilde{x}_s)$. The tower property and induction prove (8). For the stated parameters, $1 - 2L\eta = 4/5$, and the two terms in (7) are at most $1/4$ and $1/4$. $\square$

Oracle complexity and accuracy. Let $\kappa = L/\mu$ and $\Delta_0 = h(\tilde{x}_0)$. With the parameters in Theorem 1, an objective target $0 < \tau < \Delta_0$ is reached after

$$S = \left\lceil \log_2 \frac{\Delta_0}{\tau} \right\rceil \implies \mathbb{E} h(\tilde{x}_S) \leq \tau. \quad (9)$$

The total component-gradient count is at most

$$N \leq S(n + 2\lceil 50\kappa \rceil) = O\left((n + \kappa) \left[1 + \log \frac{\Delta_0}{\tau}\right]\right). \quad (10)$$

If $\Delta_0 \leq \tau$, no epochs are needed for this theoretical guarantee. Choosing S in advance uses Δ_0 or an upper bound.

The objective and stationarity criteria can also be related explicitly. Applying smoothness at $x - \nabla F(x)/L$ and using the global lower bound $F_\star$ gives

$$\|\nabla F(x)\|^2 \leq 2Lh(x). \quad (11)$$

Thus taking $\tau = \varepsilon^2/(2L)$ ensures $\mathbb{E} \|\nabla F(\tilde{x}_S)\|^2 \leq \varepsilon^2$ and, by Cauchy–Schwarz, $\mathbb{E} \|\nabla F(\tilde{x}_S)\| \leq \varepsilon$. Here τ measures objective error, while ε measures gradient norm; the distinction matters when comparing rates across methods.

Interpretation and scope. The second moment in (6) decreases as both the inner iterate and the snapshot approach the solution. This permits a fixed step size and geometric improvement without assuming that every component gradient vanishes at $x_\star$. The proof requires convex components, a strongly convex average, and the specified random snapshot rule. It does not establish the same rate for the last inner iterate or a nonconvex objective. Nonconvex SVRG analyses [20, 21] and loopless variants [18, 19] use different arguments. The exact snapshot gradient is also essential to this proof; approximating it introduces another error term.

4 Recommended reading

Variance-reduction methods. Basic guarantees [1–3]; gradient memories and control variates [4–10]; sampling and acceleration [11–19]; nonconvex and recursive estimators [20–32]; momentum and adaptive schedules [33–43]; second-order extensions [44–47]; lower bounds [48].

Compositional optimization. Two-level approximation [49–52]; finite-sum and recursive estimators [53–56]; multilevel tracking [57–60]; coupled and partially observed blocks [61–65]; projection-free methods [66–70]; statistical-model applications and federated extensions [71–74].

Sign-based optimization. Quantization, signSGD, majority voting, and error feedback [75–79]; variance-reduced sign updates and Lion [80–82]; compressed gradient tracking [83–87]. Related distributed control-variate methods include SCAFFOLD, FedPAGE, decentralized hybrid estimators, and ProxSkip [88–91].

Minimax optimization. Finite-sum saddle-point methods, matrix games, and variance-reduced extragradient [92–95]; recursive, hybrid, adaptive, and decentralized nonconvex minimax methods [96–99]; a variational-inequality treatment [100].

References

- [1] Saeed Ghadimi and Guanghui Lan. Stochastic first- and zeroth-order methods for nonconvex stochastic programming. *SIAM Journal on Optimization*, 23(4):2341–2368, 2013.
- [2] Robert M. Gower, Mark Schmidt, Francis Bach, and Peter Richtárik. Variance-reduced methods for machine learning. *Proceedings of the IEEE*, 108(11):1968–1983, 2020.
- [3] Eduard Gorbunov, Filip Hanzely, and Peter Richtárik. A Unified Theory of SGD: Variance Reduction, Sampling, Quantization and Coordinate Descent. In *AISTATS*, PMLR 108:680–690, 2020.
- [4] Nicolas Le Roux, Mark Schmidt, and Francis Bach. A stochastic gradient method with an exponential convergence rate for finite training sets. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 25, 2012.
- [5] Shai Shalev-Shwartz and Tong Zhang. Stochastic Dual Coordinate Ascent Methods for Regularized Loss Minimization. *Journal of Machine Learning Research*, 14(16):567–599, 2013.
- [6] Rie Johnson and Tong Zhang. Accelerating stochastic gradient descent using predictive variance reduction. In *NeurIPS*, volume 26, 2013.
- [7] Jakub Konecny and Peter Richtárik. Semi-Stochastic Gradient Descent Methods. *arXiv preprint arXiv:1312.1666*, 2013.
- [8] Aaron Defazio, Francis Bach, and Simon Lacoste-Julien. SAGA: A fast incremental gradient method with support for non-strongly convex composite objectives. In *NeurIPS*, volume 27, 2014.
- [9] Aaron J. Defazio, Tiberio S. Caetano, and Justin Domke. Finito: A Faster, Permutable Incremental Gradient Method for Big Data Problems. *arXiv preprint arXiv:1407.2710*, 2014.
- [10] Lin Xiao and Tong Zhang. A proximal stochastic gradient method with progressive variance reduction. *SIAM Journal on Optimization*, 24(4):2057–2075, 2014.
- [11] Hongzhou Lin, Julien Mairal, and Zaid Harchaoui. A Universal Catalyst for First-Order Optimization. In *NeurIPS*, volume 28, 2015.
- [12] Zeyuan Allen-Zhu. Katyusha: The first direct acceleration of stochastic gradient methods. *arXiv preprint arXiv:1603.05953*, 2016.

- [13] Zeyuan Allen-Zhu and Yang Yuan. Improved SVRG for Non-Strongly-Convex or Sum-of-Non-Convex Objectives. In *ICML*, PMLR 48:1080–1089, 2016.
- [14] Zeyuan Allen-Zhu. Katyusha X: Simple Momentum Method for Stochastic Sum-of-Nonconvex Optimization. In *ICML*, PMLR 80:179–185, 2018.
- [15] Xun Qian, Zheng Qu, and Peter Richtarik. SAGA with Arbitrary Sampling. In *ICML*, PMLR 97:5190–5199, 2019.
- [16] Robert M. Gower, Peter Richtarik, and Francis Bach. Stochastic Quasi-Gradient Methods: Variance Reduction via Jacobian Sketching. *arXiv preprint arXiv:1805.02632*, 2018.
- [17] Fanhua Shang, Yuanyuan Liu, Kaiwen Zhou, James Cheng, Kelvin Kai Wing Ng, and Yuichi Yoshida. Guaranteed Sufficient Decrease for Stochastic Variance Reduced Gradient Optimization. In *AISTATS*, PMLR 84:1027–1036, 2018.
- [18] Dmitry Kovalev, Samuel Horvath, and Peter Richtarik. Don’t Jump Through Hoops and Remove Those Loops: SVRG and Katyusha are Better Without the Outer Loop. *arXiv preprint arXiv:1901.08689*, 2019.
- [19] Xun Qian, Zheng Qu, and Peter Richtarik. L-SVRG and L-Katyusha with Arbitrary Sampling. *Journal of Machine Learning Research*, 22(112), 2021.
- [20] Sashank J. Reddi, Ahmed Hefny, Suvrit Sra, Barnabas Poczos, and Alex Smola. Stochastic variance reduction for nonconvex optimization. In *International Conference on Machine Learning (ICML)*, PMLR 48:314–323, 2016.
- [21] Zeyuan Allen-Zhu and Elad Hazan. Variance reduction for faster non-convex optimization. In *ICML*, PMLR 48:699–707, 2016.
- [22] Zeyuan Allen-Zhu. Natasha: Faster Non-Convex Stochastic Optimization via Strongly Non-Convex Parameter. In *ICML*, PMLR 70:89–97, 2017.
- [23] Lam M. Nguyen, Jie Liu, Katya Scheinberg, and Martin Takáč. SARAH: A novel method for machine learning problems using stochastic recursive gradient. In *ICML*, 2017.
- [24] Lam M. Nguyen, Katya Scheinberg, and Martin Takac. Inexact SARAH Algorithm for Stochastic Optimization. *arXiv preprint arXiv:1811.10105*, 2018.
- [25] Lihua Lei, Cheng Ju, Jianbo Chen, and Michael I. Jordan. Non-convex finite-sum optimization via SCSSG methods. In *NeurIPS*, volume 30, 2017.
- [26] Cong Fang, Chris Junchi Li, Zhouchen Lin, and Tong Zhang. SPIDER: Near-optimal non-convex optimization via stochastic path-integrated differential estimator. In *NeurIPS*, volume 31, 2018.
- [27] Dongruo Zhou, Pan Xu, and Quanquan Gu. Stochastic nested variance reduction for nonconvex optimization. In *NeurIPS*, volume 31, 2018.
- [28] Zhe Wang, Kaiyi Ji, Yi Zhou, Yingbin Liang, and Vahid Tarokh. SpiderBoost and momentum: Faster stochastic variance reduction algorithms. In *NeurIPS*, volume 32, 2019.
- [29] Nhan H. Pham, Lam M. Nguyen, Dzong T. Phan, and Quoc Tran-Dinh. ProxSARAH: An Efficient Algorithmic Framework for Stochastic Composite Nonconvex Optimization. *Journal of Machine Learning Research*, 21(110):1–48, 2020.
- [30] Zhize Li, Slavomir Hanzely, and Peter Richtarik. ZeroSARAH: Efficient Nonconvex Finite-Sum Optimization with Zero Full Gradient Computation. *arXiv preprint arXiv:2103.01447*, 2021.
- [31] Zhize Li, Hongyan Bao, Xiangliang Zhang, and Peter Richtárik. PAGE: A simple and optimal probabilistic gradient estimator for nonconvex optimization. In *ICML*, 2021.
- [32] Samuel Horvath and Peter Richtarik. Nonconvex Variance Reduced Optimization with Arbitrary Sampling. In *ICML*, PMLR 97:2781–2789, 2019.

- [33] Ashok Cutkosky and Francesco Orabona. Momentum-based variance reduction in non-convex SGD. In *NeurIPS*, volume 32, 2019.
- [34] Quoc Tran-Dinh, Nhan H. Pham, Dzung T. Phan, and Lam M. Nguyen. A hybrid stochastic optimization framework for stochastic composite nonconvex optimization. *arXiv preprint arXiv:1907.03793*, 2019.
- [35] Kfir Y. Levy, Ali Kavis, and Volkan Cevher. STORM+: Fully adaptive SGD with recursive momentum for nonconvex optimization. In *NeurIPS*, volume 34, 2021.
- [36] Feihu Huang, Junyi Li, and Heng Huang. SUPER-ADAM: Faster and Universal Framework of Adaptive Gradients. In *NeurIPS*, volume 34, 2021.
- [37] Zijian Liu, Ta Duy Nguyen, Thien Hang Nguyen, Alina Ene, and Huy L. Nguyen. META-STORM: Generalized fully-adaptive variance reduced SGD for unbounded functions. *arXiv preprint arXiv:2209.14853*, 2022.
- [38] Ali Kavis, Stratis Skoulakis, Kimon Antonakopoulos, Leello Tadesse Dadi, and Volkan Cevher. Adaptive Stochastic Variance Reduction for Non-convex Finite-Sum Minimization. In *NeurIPS*, volume 35, 2022.
- [39] Wei Jiang, Sifan Yang, Yibo Wang, and Lijun Zhang. Adaptive variance reduction for stochastic optimization under weaker assumptions. In *NeurIPS*, volume 37, pages 22047–22080, 2024.
- [40] Benjamin Dubois-Taine, Sharan Vaswani, Reza Babanezhad, Mark Schmidt, and Simon Lacoste-Julien. SVRG Meets AdaGrad: Painless Variance Reduction. *arXiv preprint arXiv:2102.09645*, 2021.
- [41] Bingcong Li, Lingda Wang, and Georgios B. Giannakis. Almost Tune-Free Variance Reduction. In *ICML*, PMLR 119:5969–5978, 2020.
- [42] Zijian Liu, Ta Duy Nguyen, Alina Ene, and Huy L. Nguyen. Adaptive Accelerated (Extra-)Gradient Methods with Variance Reduction. In *ICML*, PMLR 162:13947–13994, 2022.
- [43] Ganzhao Yuan. Adaptive Lipschitz-Free Proximal Gradient Methods for Stochastic Composite Nonconvex Optimization. *arXiv preprint arXiv:2502.21099v4*, 2026.
- [44] Zhize Li. SSRGD: Simple Stochastic Recursive Gradient Descent for Escaping Saddle Points. In *NeurIPS*, volume 32, 2019.
- [45] Zeyuan Allen-Zhu. Natasha 2: Faster Non-Convex Optimization Than SGD. In *NeurIPS*, volume 31, 2018.
- [46] Zhe Wang, Yi Zhou, Yingbin Liang, and Guanghui Lan. Stochastic Variance-Reduced Cubic Regularization for Nonconvex Optimization. In *AISTATS*, PMLR 89:2731–2740, 2019.
- [47] Dongruo Zhou, Pan Xu, and Quanquan Gu. Stochastic Variance-Reduced Cubic Regularized Newton Methods. In *ICML*, PMLR 80:5990–5999, 2018.
- [48] Yossi Arjevani, Yair Carmon, John C. Duchi, Dylan J. Foster, Nathan Srebro, and Blake Woodworth. Lower bounds for non-convex stochastic optimization. *Mathematical Programming*, 199:165–214, 2023.
- [49] Mengdi Wang, Ethan X. Fang, and Han Liu. Stochastic compositional gradient descent: Algorithms for minimizing compositions of expected-value functions. *arXiv preprint arXiv:1411.3803*, 2014.
- [50] Mengdi Wang, Ji Liu, and Ethan X. Fang. Accelerating stochastic composition optimization. *Journal of Machine Learning Research*, 18(105):1–23, 2017.
- [51] Saeed Ghadimi, Andrzej Ruszczyński, and Mengdi Wang. A Single Timescale Stochastic Approximation Method for Nested Stochastic Optimization. *SIAM Journal on Optimization*, 2020.
- [52] Tianyi Chen, Yuejiao Sun, and Wotao Yin. Solving Stochastic Compositional Optimization is Nearly as Easy as Solving Stochastic Optimization. *arXiv preprint arXiv:2008.10847*, 2020.

- [53] Xiangru Lian, Mengdi Wang, and Ji Liu. Finite-sum composition optimization via variance reduced gradient descent. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, PMLR 54:1159–1167, 2017.
- [54] Tsung-Yu Hsieh, Yasser El-Manzalawy, Yiwei Sun, and Vasant Honavar. Compositional Stochastic Average Gradient for Machine Learning and Related Applications. *arXiv preprint arXiv:1809.01225*, 2018.
- [55] Junyu Zhang and Lin Xiao. A composite randomized incremental gradient method. In *ICML*, PMLR 97:7454–7462, 2019.
- [56] Wenqing Hu, Chris Junchi Li, Xiangru Lian, Ji Liu, and Huizhuo Yuan. Efficient Smooth Non-Convex Stochastic Compositional Optimization via Stochastic Recursive Gradient Descent. In *NeurIPS*, volume 32, 2019.
- [57] Junyu Zhang and Lin Xiao. Multi-level composite stochastic optimization via nested variance reduction. *arXiv preprint arXiv:1908.11468*, 2019.
- [58] Wei Jiang, Bokun Wang, Yibo Wang, Lijun Zhang, and Tianbao Yang. Optimal algorithms for stochastic multi-level compositional optimization. In *ICML*, PMLR 162:10195–10216, 2022.
- [59] Krishnakumar Balasubramanian, Saeed Ghadimi, and Anthony Nguyen. Stochastic Multi-level Composition Optimization Algorithms with Level-Independent Convergence Rates. *arXiv preprint arXiv:2008.10526*, 2020.
- [60] Wei Jiang, Sifan Yang, Yibo Wang, Tianbao Yang, and Lijun Zhang. Revisiting stochastic multi-level compositional optimization. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 47(7):5613–5624, 2025.
- [61] Bokun Wang and Tianbao Yang. Finite-Sum Coupled Compositional Stochastic Optimization: Theory and Applications. In *ICML*, PMLR 162:23292–23317, 2022.
- [62] Wei Jiang, Gang Li, Yibo Wang, Lijun Zhang, and Tianbao Yang. Multi-block-single-probe variance reduced estimator for coupled compositional optimization. In *NeurIPS*, volume 35, pages 32499–32511, 2022.
- [63] Quanqi Hu, Dixian Zhu, and Tianbao Yang. Non-Smooth Weakly-Convex Finite-sum Coupled Compositional Optimization. In *NeurIPS*, volume 36, 2023.
- [64] Wei Jiang, Sifan Yang, Yibo Wang, Lijun Zhang, and Zechao Li. Solving finite-sum coupled compositional optimization via multi-block-single-probe estimator. *arXiv preprint arXiv:2609.15723*, 2026.
- [65] Xingyu Chen, Bokun Wang, Ming Yang, Qihang Lin, and Tianbao Yang. Stochastic Momentum Methods for Non-smooth Non-Convex Finite-Sum Coupled Compositional Optimization. In *NeurIPS*, volume 38, 2025.
- [66] Elad Hazan and Haipeng Luo. Variance-Reduced and Projection-Free Stochastic Optimization. In *ICML*, PMLR 48:1263–1271, 2016.
- [67] Tesi Xiao, Krishnakumar Balasubramanian, and Saeed Ghadimi. A Projection-free Algorithm for Constrained Stochastic Multi-level Composition Optimization. In *NeurIPS*, volume 35, 2022.
- [68] Wei Jiang, Sifan Yang, Wenhao Yang, Yibo Wang, Yuanyu Wan, and Lijun Zhang. Projection-free variance reduction methods for stochastic constrained multi-level compositional optimization. In *ICML*, PMLR 235:21962–21987, 2024.
- [69] Hamed Hassani, Amin Karbasi, Aryan Mokhtari, and Zebang Shen. Stochastic Conditional Gradient++. *arXiv preprint arXiv:1902.06992*, 2019.
- [70] Wei Jiang, Sifan Yang, Wenhao Yang, Yibo Wang, Yuanyu Wan, Zechao Li, and Lijun Zhang. Projection-free multi-level algorithms for stochastic constrained compositional optimization. *arXiv preprint arXiv:2609.15679*, 2026.

- [71] Wei Jiang, Jiayu Qin, Lingyu Wu, Changyou Chen, Tianbao Yang, and Lijun Zhang. Learning unnormalized statistical models via compositional optimization. In *ICML*, PMLR 202:15105–15124, 2023.
- [72] Davoud Ataee Tarzanagh, Mingchen Li, Christos Thrampoulidis, and Samet Oymak. FedNest: Federated Bilevel, Minimax, and Compositional Optimization. In *ICML*, PMLR 162:21146–21179, 2022.
- [73] Wei Jiang, Jiayu Qin, Lingyu Wu, Changyou Chen, Tianbao Yang, and Lijun Zhang. Optimizing unnormalized statistical models through compositional optimization. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 48(2):1949–1960, 2026.
- [74] Hongchang Gao, Junyi Li, and Heng Huang. On the Convergence of Local Stochastic Compositional Gradient Descent with Momentum. In *ICML*, PMLR 162:7017–7035, 2022.
- [75] Dan Alistarh, Demjan Grubic, Jerry Li, Ryota Tomioka, and Milan Vojnovic. QSGD: Communication-Efficient SGD via Gradient Quantization and Encoding. In *NeurIPS*, volume 30, 2017.
- [76] Wei Wen, Cong Xu, Feng Yan, Chunpeng Wu, Yandan Wang, Yiran Chen, and Hai Li. TernGrad: Ternary Gradients to Reduce Communication in Distributed Deep Learning. In *NeurIPS*, volume 30, 2017.
- [77] Jeremy Bernstein, Yu-Xiang Wang, Kamyar Azizzadenesheli, and Anima Anandkumar. signSGD: Compressed optimisation for non-convex problems. In *ICML*, PMLR 80:560–569, 2018.
- [78] Jeremy Bernstein, Jiawei Zhao, Kamyar Azizzadenesheli, and Anima Anandkumar. signSGD with majority vote is communication efficient and fault tolerant. In *International Conference on Learning Representations (ICLR)*, 2019.
- [79] Sai Praneeth Karimireddy, Quentin Rebjock, Sebastian U. Stich, and Martin Jaggi. Error feedback fixes SignSGD and other gradient compression schemes. In *ICML*, PMLR 97:3252–3261, 2019.
- [80] Wei Jiang, Sifan Yang, Wenhao Yang, and Lijun Zhang. Efficient sign-based optimization: Accelerating convergence via variance reduction. In *NeurIPS*, volume 37, pages 33891–33932, 2024.
- [81] Xiangning Chen, Chen Liang, Da Huang, Esteban Real, Kaiyuan Wang, Hieu Pham, Xuanyi Dong, Thang Luong, Cho-Jui Hsieh, Yifeng Lu, and Quoc V. Le. Symbolic discovery of optimization algorithms. In *NeurIPS*, volume 36, 2023.
- [82] Wei Jiang, Mao Xu, Wenhao Yang, Yibo Wang, Zechao Li, and Lijun Zhang. Convergence analysis of the Lion optimizer in centralized and distributed settings. In *ICML*, PMLR 306:53308–53333, 2026.
- [83] Konstantin Mishchenko, Eduard Gorbunov, Martin Takac, and Peter Richtarik. Distributed Learning with Compressed Gradient Differences. *arXiv preprint arXiv:1901.09269*, 2019.
- [84] Eduard Gorbunov, Konstantin P. Burlachenko, Zhize Li, and Peter Richtarik. MARINA: Faster Non-Convex Distributed Learning with Compression. In *ICML*, PMLR 139:3788–3798, 2021.
- [85] Wei Jiang, Zechao Li, and Lijun Zhang. Revisiting distributed sign-based variance reduction. *arXiv preprint arXiv:2609.18656*, 2026.
- [86] Peter Richtarik, Igor Sokolov, and Ilyas Fatkhullin. EF21: A New, Simpler, Theoretically Better, and Practically Faster Error Feedback. In *NeurIPS*, volume 34, 2021.
- [87] Alexander Tyurin and Peter Richtarik. DASHA: Distributed Nonconvex Optimization with Communication Compression, Optimal Oracle Complexity, and No Client Synchronization. *arXiv preprint arXiv:2202.01268*, 2022.
- [88] Sai Praneeth Karimireddy, Satyen Kale, Mehryar Mohri, Sashank Reddi, Sebastian Stich, and Ananda Theertha Suresh. SCAFFOLD: Stochastic Controlled Averaging for Federated Learning. In *ICML*, PMLR 119:5132–5143, 2020.

- [89] Haoyu Zhao, Zhize Li, and Peter Richtarik. FedPAGE: A Fast Local Stochastic Gradient Method for Communication-Efficient Federated Learning. *arXiv preprint arXiv:2108.04755*, 2021.
- [90] Ran Xin, Usman A. Khan, and Soumya Kar. A Hybrid Variance-Reduced Method for Decentralized Stochastic Non-Convex Optimization. In *ICML*, PMLR 139:11459–11469, 2021.
- [91] Konstantin Mishchenko, Grigory Malinovsky, Sebastian Stich, and Peter Richtarik. ProxSkip: Yes! Local Gradient Steps Provably Lead to Communication Acceleration! Finally!. In *ICML*, PMLR 162:15750–15769, 2022.
- [92] P. Balamurugan and Francis Bach. Stochastic variance reduction methods for saddle-point problems. In *NeurIPS*, volume 29, 2016.
- [93] Yair Carmon, Yujia Jin, Aaron Sidford, and Kevin Tian. Variance Reduction for Matrix Games. In *NeurIPS*, volume 32, 2019.
- [94] Tatjana Chavdarova, Gauthier Gidel, Francois Fleuret, and Simon Lacoste-Julien. Reducing Noise in GAN Training with Variance Reduced Extragradient. In *NeurIPS*, volume 32, 2019.
- [95] Junchi Yang, Siqi Zhang, Negar Kiyavash, and Niao He. A Catalyst Framework for Minimax Optimization. In *NeurIPS*, volume 33, 2020.
- [96] Luo Luo, Haishan Ye, Zhichao Huang, and Tong Zhang. Stochastic recursive gradient descent ascent for stochastic nonconvex-strongly-concave minimax problems. In *NeurIPS*, volume 33, 2020.
- [97] Quoc Tran-Dinh, Deyi Liu, and Lam M. Nguyen. Hybrid Variance-Reduced SGD Algorithms for Minimax Problems with Nonconvex-Linear Function. In *NeurIPS*, volume 33, 2020.
- [98] Feihu Huang, Xidong Wu, and Zhengmian Hu. AdaGDA: Faster adaptive gradient descent ascent methods for minimax optimization. In *AISTATS*, PMLR 206:2365–2389, 2023.
- [99] Lesi Chen, Haishan Ye, and Luo Luo. An Efficient Stochastic Algorithm for Decentralized Nonconvex-Strongly-Concave Minimax Optimization. In *AISTATS*, PMLR 238:1990–1998, 2024.
- [100] Ahmet Alacaoglu and Yura Malitsky. Stochastic variance reduction for variational inequality methods. In *Conference on Learning Theory (COLT)*, PMLR 178:778–816, 2022.